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On the threshold for rainbow connection number r in random graphs

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Abstract

We call an edge colouring of a graph G a *rainbow colouring* if every pair of vertices is joined by a *rainbow path*, i.e., a path where no two edges have the same colour. The minimum number of colours required for a rainbow colouring of the edges of G is called the *rainbow connection number* (or *rainbow connectivity*) $\mathrm{rc}(G)$ of G. We investigate sharp thresholds in the Erdős–Rényi random graph for the property "$\mathrm{rc}(G) \leqslant r$" where r is a fixed integer. It is known that for $r = 2$, rainbow connection number 2 and diameter 2 happen essentially at the same time in random graphs. For $r \geqslant 3$, we conjecture that this is not the case, propose an alternative threshold, and prove that this is an upper bound for the threshold for rainbow connection number r.

1 Introduction

The rainbow connection number is a measure of the connectivity of a graph introduced in 2008 by Chartrand, Johns, McKeon and Zhang [3], which has recently attracted the attention of a number of researchers (see [9]).

An edge colouring of a graph G is called a *rainbow colouring* if every pair of vertices is joined by a *rainbow path*, i.e., a path where no two edges have the same colour. The minimum number of colours required for a rainbow colouring of the edges of G is called the *rainbow connection number* (or *rainbow connectivity*) $\mathrm{rc}(G)$ of G.

In this paper we investigate the rainbow connection numbers of random graphs. As usual, we say that an event $E = E(n)$ holds *with high probability*

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(whp) if $\lim_{n\to\infty}\mathbb{P}(E) = 1$. For two functions $f, g : \mathbb{N} \to \mathbb{R}$, we write $f = O(g)$ if there are constants C and n_0 such that $|f(n)| \leqslant Cg(n)$ for all $n \geqslant n_0$, and $f = o(g)$ if $f(n)/g(n) \to 0$ as $n \to \infty$. Furthermore, we say $f = \Theta(g)$ if $f = O(g)$ and $g = O(f)$. Finally, we write $f = O^*(g)$ if there are constants C and n_0 such that $|f(n)| \leqslant (\log n)^C g(n)$ for all $n \geqslant n_0$, where $\log n$ denotes the natural logarithm.

Recall that the *Erdős–Rényi random graph* with edge probability $p = p(n)$, denoted by $G \sim \mathcal{G}(n,p)$, is a graph with n vertices where each of the $\binom{n}{2}$ possible edges is included with probability p, independently of the others. A sequence $p^* = p^*(n)$ is called a *sharp threshold* for a graph property $\mathcal{P}$ if for all constants $c < 1, C > 1$, if $p(n) < cp^*(n)$ then whp $\mathcal{G}(n,p) \notin \mathcal{P}$, and if $p(n) > Cp^*(n)$ then whp $\mathcal{G}(n,p) \in \mathcal{P}$. Recall that for a (weak) threshold, the conditions assume $p = o(p^*)$ and $p^* = o(p)$, respectively. We also use the non-standard notion of a *semisharp threshold*, which only requires the existence of some constants $c, C > 0$ such that the properties above hold.

We are interested in the graph property

$$\mathcal{R}_r = \{G : \mathrm{rc}(G) \leqslant r\}$$

of having rainbow connection number at most r. Caro, Lev, Roditty, Tuza and Yuster [2] showed that $\sqrt{\frac{\log n}{n}}$ is a semisharp threshold for $\mathcal{R}_2$, and He and Liang [6] proved that for general $r \geqslant 2$, $\frac{(\log n)^{1/r}}{n^{1-1/r}}$ is a semi-sharp threshold for $\mathcal{R}_r$. As observed by Friedgut [5], a coarse threshold can only occur near rational powers of n. From Theorem 1.4 in [5], if the semi-sharp threshold for $\mathcal{R}_r$ were not sharp, there would be a sequence (n_k) and $p(n_k) = \Theta\left(\frac{(\log n_k)^{1/r}}{n_k^{1-1/r}}\right)$ such that $b_1 n_k^\alpha \leqslant p(n_k) \leqslant b_2 n_k^\alpha$ for some constants $b_1, b_2 \in \mathbb{R}$ and $\alpha \in \mathbb{Q}$, a contradiction. Hence, a *sharp threshold* for the property $\mathcal{R}_r$ exists.

Bollobás [1] showed that for any fixed $r \geqslant 2$, $\frac{(2\log n)^{1/r}}{n^{1-1/r}}$ is a sharp threshold for the graph property

$$\mathcal{D}_r = \{G : \mathrm{diam}(G) \leqslant r\}$$

of having diameter at most r. In particular, $\mathcal{R}_r$ and $\mathcal{D}_r$ have the same weak threshold. It is an easy observation that for any connected graph G, $\mathrm{rc}(G) \geqslant \mathrm{diam}(G)$, and therefore $\frac{(2\log n)^{1/r}}{n^{1-1/r}}$ is a lower bound for the sharp threshold of $\mathcal{R}_r$.

In [7], we showed that for $r = 2$, the hitting times of the properties $\mathcal{R}_2$ and $\mathcal{D}_2$ coincide whp, so in particular these properties have the same

sharp threshold. It is a natural question to ask whether this result may be extended to $r \geqslant 3$. However, it seems that the situation for $r \geqslant 3$ is fundamentally different from the case $r = 2$, and the methods used in [7] do not carry over to the general case. In fact, there are good reasons to believe that the following may be the true sharp threshold for $\mathcal{R}_r$ where $r \geqslant 3$.

Conjecture 1. *Fix an integer $r \geqslant 3$, set $C = \frac{r^{r-2}}{(r-2)!}$, and let*

$$p(n) = \frac{(C \log n)^{1/r}}{n^{1-1/r}}. \tag{1}$$

Then $p(n)$ is a sharp threshold for the graph property $\mathcal{R}_r$.

In one direction, there is a heuristic argument that (1) is a lower bound for the sharp threshold for $\mathcal{R}_r$. Let $\varepsilon > 0$ and $p = \frac{(C(1-\varepsilon)\log n)^{1/r}}{n^{1-1/r}}$, and colour the edges of $G \sim \mathcal{G}(n,p)$ with r colours independently and uniformly at random. For a given pair of vertices v, w, the expected number of rainbow paths joining them is about $\frac{r!}{r^r} n^{r-1} p^r = (1-\varepsilon)\left(1 - \frac{1}{r}\right) \log n$, and the probability that v and w are not joined by any rainbow path is about $n^{-(1-\varepsilon)\left(1-\frac{1}{r}\right)}$. The expected number of pairs of vertices not joined by any rainbow path is therefore $\Theta\left(n^{1+\frac{1}{r}+\varepsilon\left(1-\frac{1}{r}\right)}\right)$. Assuming that these events behave roughly independently for different pairs of vertices, we would expect the overall probability of the random colouring being a rainbow colouring to be about $\exp\left(-\Theta\left(n^{1+\frac{1}{r}+\varepsilon\left(1-\frac{1}{r}\right)}\right)\right)$. Conditional on G having $O^*\left(n^{1+\frac{1}{r}}\right)$ edges, which holds with very high probability, there are $\exp\left(O^*\left(n^{1+\frac{1}{r}}\right)\right)$ possible edge colourings. The probability that there exists at least one rainbow colouring is then bounded by the total number of colourings multiplied with the probability that a random colouring is a rainbow colouring, which tends to 0.

The best known upper bound for the sharp threshold of the property $\mathcal{R}_r$ is $\frac{(2^{20r}\log n)^{1/r}}{n^{1-1/r}}$, which was shown in the proof of the semi-sharp threshold in [6]. We will establish the other direction of Conjecture 1 in this article, i.e., we will prove the following result.

Theorem 2. *Fix an integer $r \geqslant 3$ and $\varepsilon > 0$. Set $p = p(n) = \frac{(C(1+\varepsilon)\log n)^{1/r}}{n^{1-1/r}}$, and let $G \sim \mathcal{G}(n,p)$. Then whp,* $\mathrm{rc}(G) = r$.

2 Proof of Theorem 2

Let $G = (V, E) \sim \mathcal{G}(n,p)$. The basic idea of the proof is as follows. First we colour the edges of G independently and uniformly at random using r

colours. We call a pair of vertices *dangerous* if it is joined by at most K rainbow paths of length r in this colouring, where K is a constant which will be defined later.

For each dangerous pair, we will select one path joining it and change the colours of the edges to make it a rainbow path, which will yield a rainbow colouring. To see that this is possible without any conflicts and that this does not rainbow-disconnect the pairs that previously had many rainbow paths, we need to study the structure of the graph and its dangerous pairs.

The proof is organised as follows. In Section 2.1 some of the known bounds needed for our later work will be reviewed. Section 2.2 contains general observations on the distribution of edges of each colour and paths of length r in the randomly coloured graph. The heart of the proof is Section 2.3, where the key lemma is proved. This lemma ensures that when we later select a path of length r for every dangerous pair of vertices and recolour it to make it a rainbow path, it is possible to do so without using any edges from a path that was previously assigned to another dangerous pair of vertices. Finally, in Section 2.4, the recolouring procedure will be described in detail and it will be shown that we can indeed find a rainbow r-colouring of the edges of G with this strategy.

2.1 Preliminaries

We will need a number of bounds for the tail of binomial distributions, which are derived from the well-known Chernoff bounds ([4], see also [8, p.26]):

Lemma 3. *Let X be distributed binomially with parameters n and p, and let $0 < x < 1$.*

(i) If $x \geqslant p$, then $\mathbb{P}(X \geqslant nx) \leqslant \left[\left(\frac{p}{x}\right)^x \left(\frac{1-p}{1-x}\right)^{1-x}\right]^n$.

(ii) If $x \leqslant p$, then $\mathbb{P}(X \leqslant nx) \leqslant \left[\left(\frac{p}{x}\right)^x \left(\frac{1-p}{1-x}\right)^{1-x}\right]^n$.

□

The following corollary of the Chernoff bounds is given in Theorem 2.1 in [8].

Corollary 4. *Let X be distributed binomially with parameters n and p, and let $\varphi(x) = (1+x)\log(1+x) - x$ for $x \geqslant -1$ and $\varphi(x) = -\infty$ otherwise. Then for all $t \geqslant 0$,*

$$\mathbb{P}(X \leqslant np - t) \leqslant \exp\left(-np\varphi\left(-\frac{t}{np}\right)\right).$$